Connections : Semester 1, Week 1.

Bertsekas and Tsitsiklis Probability Book, Chapter 1

1.4 (a) B: Sum of two rolls ≤ 4

A: doubles are rolled

A : $\{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$

B: $\{\underbrace{(1,1)}_{\text{sum}=2}, \underbrace{(1,2), (2,1)}_{\text{Sum}=3}, \underbrace{(2,2), (1,3), (3,1)}_{\text{Sum}=4}\}$

$\Rightarrow A \cap B = \{(1,1), (2,2)\}$

$\Rightarrow P(A|B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{|A \cap B|}{|B|} = \frac{1}{3}$.

1.5 Let A : two heads in two coin tosses = $\{HH\}$

B: first toss is a head = $\{HH, HT\}$

C : at least one toss is a head = $\{HH, HT, TH\}$

Clearly $B \subseteq C \Rightarrow P(B) < P(C)$ and $A \cap B = A \cap C$

$\Rightarrow P(A \cap B) = P(A \cap C)$

$\Rightarrow P(A|B) = \dfrac{P(A \cap B)}{P(B)} > \dfrac{P(A \cap C)}{P(C)} = P(A|C)$

7 The sample space will be the set of outcomes

$\{ o_1 o_2 o_3 \cdots o_n e \mid o_i \in \{1,3\}, e \in \{2,4\}, n \in \mathbb{N} \}$

An outcome is a sequence of numbers till an even number is obtained (including it). It also includes infinite sequences of odd numbers.

8 Let W_i = event of winning against ith opponent

L_i = event of losing against ith opponent

Favorable outcomes are only : $W_1W_2W_3$, $W_1W_2L_3$, $L_1W_2W_3$

Let p_i = probability of W_i $\Rightarrow 1-p_i$ = probability of L_i

$\Rightarrow P(\text{Win}) = P(\{W_1W_2W_3, W_1W_2L_3, L_1W_2W_3\})$

$= p_1p_2 + (1-p_1)p_2p_3 = p_2(p_1 + p_3 - p_1p_3)$

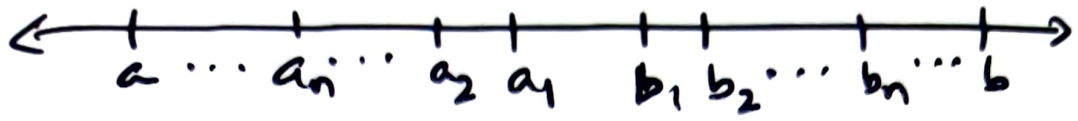
Since the expression is symmetric in p_1 and p_3, we only need to consider the cases

$p_1 > p_2 > p_3$; $p_1 > p_3 > p_2$; and $p_2 > p_1 > p_3$

How to show that $P(\text{win})$ is the greatest for $p_1 > p_3 > p_2$?

Problem Set 1

G1(b)



$\lim_{n\to\infty} a_n = a$ and $\lim_{n\to\infty} b_n = b$

Let $A_n = [a_n, b_n] \Rightarrow A_{n-1} \subset A_n \quad \forall n \in \mathbb{N}$

Let $B_n = A_n \cap A_{n-1}^c = [a_n, a_{n-1}) \cup (b_{n-1}, b_n]$

$\Rightarrow A_n = \bigcup_{k=1}^{n} B_k$ and $B_i \cap B_j = \emptyset \quad \forall i, j \in \mathbb{N}$

$\Rightarrow A = \lim_{n\to\infty} A_n = \lim_{n\to\infty} \bigcup_{k=1}^{n} B_k \qquad [A = (a,b)]$

$\Rightarrow P(A) = \lim_{n\to\infty} P\left(\bigcup_{k=1}^{n} B_k\right) = \lim_{n\to\infty} P(A_n)$

$\Rightarrow \lim_{n\to\infty} P([a_n, b_n]) = P((a,b))$

Mathematics of Machine Learning Chapters 2 and 3 test out

2.1 (b) $3 * x * x = 15$

$\Rightarrow 3 * (x^2 + 2x) = 15$

$\Rightarrow 3x^2 + 6x + x^2 + 2x + 3 = 15$

$\Rightarrow 4x^2 + 8x = 12 \Rightarrow x^2 + 2x - 3 = 0 \Rightarrow x = 1, -3$

2.3 Let $A = \begin{bmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$,

$C = \begin{bmatrix} 1 & p & q \\ 0 & 1 & r \\ 0 & 0 & 1 \end{bmatrix} \in G$

$A(BC) = \begin{bmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & a+p & q+ar+b \\ 0 & 1 & r+c \\ 0 & 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1 & a+p+x & q+b+z+ar+rx+cx \\ 0 & 1 & c+r+y \\ 0 & 0 & 1 \end{bmatrix}$

$(AB)C = \begin{bmatrix} 1 & a+x & b+cx+z \\ 0 & 1 & c+y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & p & q \\ 0 & 1 & r \\ 0 & 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1 & a+p+x & q+b+z+ar+rx+cx \\ 0 & 1 & c+r+y \\ 0 & 0 & 1 \end{bmatrix} = A(BC)$

$\Rightarrow \cdot$ is associative on G

$\Rightarrow I$ is the identity element

$$\forall A = \begin{bmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{bmatrix}, \quad \tilde{A} = \begin{bmatrix} 1 & -x & xy-z \\ 0 & 1 & -y \\ 0 & 0 & 1 \end{bmatrix} \in G$$

Such that $\tilde{A}A = A\tilde{A} = I$

$\Rightarrow$ inverse exists in G for $\cdot$

$$AB = \begin{bmatrix} 1 & a+x & b+cx+z \\ 0 & 1 & c+y \\ 0 & 0 & 1 \end{bmatrix} \quad BA = \begin{bmatrix} 1 & x+a & z+ay+b \\ 0 & 1 & y+c \\ 0 & 0 & 1 \end{bmatrix}$$

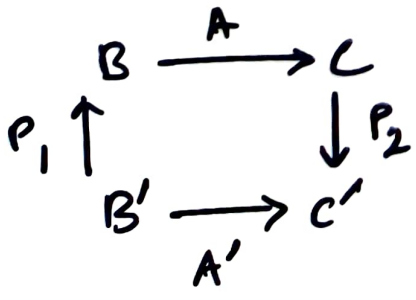
$AB \neq BA \Rightarrow \cdot$ is not commutative in G

$AB \in G \Rightarrow \cdot$ satisfies closure on G

$\Rightarrow G = (G, \cdot)$ is a group but it is not Abelian

2.20 $P_1 = B^{-1}B' = \begin{bmatrix} 4 & 0 \\ 6 & -1 \end{bmatrix}$ and $P_2 = C = \begin{bmatrix} 1 & 0 & 1 \\ 2 & -1 & 0 \\ -1 & 2 & -1 \end{bmatrix}$

$P_1 : B' \to B$ and $P_2 : C \to C'$



(a) $\phi(b_1 + b_2) = c_2 + c_3$

$\phi(b_1 - b_2) = 2c_1 - c_2 + 3c_3$

$\Rightarrow \phi(b_1) = c_1 + 2c_3$

$\phi(b_2) = -c_1 + c_2 - c_3$

$$\Rightarrow A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 2 & -1 \end{bmatrix}$$

(e) $A' = P_2 A P_1 = \begin{bmatrix} 1 & 0 & 1 \\ 2 & -1 & 0 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 6 & -1 \end{bmatrix}$

$$= \begin{bmatrix} 3 & -2 \\ 2 & -3 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 6 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -10 & 3 \\ 12 & -4 \end{bmatrix}$$

(f) $x_{B'} = [2, 3]^T$

(i) $x_B = P_1 x_{B'} = \begin{bmatrix} 4 & 0 \\ 6 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 8 \\ 9 \end{bmatrix}$

(ii) $\phi(x)|_e = A x_B = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 8 \\ 9 \end{bmatrix} = \begin{bmatrix} -1 \\ 9 \\ 7 \end{bmatrix}$

(iii) $\phi(x)|_{e'} = P_2 A x_B = \begin{bmatrix} 1 & 0 & 1 \\ 2 & -1 & 0 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 9 \\ 7 \end{bmatrix} = \begin{bmatrix} 6 \\ -11 \\ 12 \end{bmatrix}$

(iv) $\phi(x)|_{e'} = A' x_{B'} = \begin{bmatrix} 0 & 2 \\ -10 & 3 \\ 12 & -4 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ -11 \\ 12 \end{bmatrix}$

3.7 (b) If $v \in \text{Im}(id_V - \pi) \Rightarrow \exists\, u \in V$ s.t. $(id_V - \pi)u = v$

$\Rightarrow \pi v = \pi(id_V - \pi)u = 0 \Rightarrow v \in \ker(\pi)$

$\Rightarrow \text{Im}(id_V - \pi) \subset \ker(\pi)$

If $v \in \ker(\pi) \Rightarrow \pi v = 0 \Rightarrow (id_V - \pi)v = v \Rightarrow v \in \operatorname{Im}(id_V - \pi)$

$\Rightarrow \ker(\pi) \subset \operatorname{Im}(id_V - \pi)$

$\Rightarrow \operatorname{Im}(id_V - \pi) = \ker(\pi)$

If $v \in \ker(id_V - \pi) \Rightarrow (id_V - \pi)v = 0 \Rightarrow \pi v = v \Rightarrow v \in \operatorname{Im}(\pi)$

$\Rightarrow \ker(id_V - \pi) \subset \operatorname{Im}(\pi)$

If $v \in \operatorname{Im}(\pi) \Rightarrow \pi v = v \Rightarrow (id_V - \pi)v = 0 \Rightarrow v \in \ker(id_V - \pi)$

$\Rightarrow \operatorname{Im}(\pi) \subset \ker(id_V - \pi)$

$\Rightarrow \ker(id_V - \pi) = \operatorname{Im}(\pi)$